Matrix A	Elementary Transformation
$A = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{pmatrix}$	
$A \sim \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 1 & 1 & 3 \end{pmatrix}$	$R_1 \leftrightarrow R_2$
$\sim \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & -3 \end{pmatrix}$	$R_3 \rightarrow R_3 - 3R_1$
$\sim \begin{pmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{pmatrix}$	$R_3 \rightarrow R_3 + 5R_2$

The number of non zero rows is 3.
$\therefore \rho(A) = 3.$

Example 1.8

Find the rank of the matrix $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 3 & 4 & 5 & 2 \\ 2 & 3 & 4 & 0 \end{pmatrix}$

Solution:

The order of A is 3×4.

$\therefore \rho(A) \leq 3.$

Let us transform the matrix A to an echelon form

Matrix A	Elementary Transformation
$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 3 & 4 & 5 & 2 \\ 2 & 3 & 4 & 0 \end{pmatrix}$	
$\sim \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -2 \end{pmatrix}$	$R_2 \rightarrow R_2 - 3R_1$ $R_3 \rightarrow R_3 - 2R_1$
$\sim \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	$R_3 \rightarrow R_3 - R_2$

The number of non zero rows is 3.
$\therefore \rho(A) = 3.$

Consistency of Equations

System of linear equations in two variables

We have already studied , how to solve two simultaneous linear equations by matrix inversion method.

Recall

Linear equations can be written in matrix form AX=B, then the solution is $X = A^{-1}B$, provided $|A| \neq 0$

Consider a system of linear equations with two variables,

$$\left.\begin{aligned} ax + by &= h \\ cx + dy &= k \end{aligned}\right\} \quad (1)$$

Where a, b, c, d, h and k are real constants and neither a and b nor c and d are both zero.

For any tow given lines L_1 and L_2, one and only one of the following may occur.

L_1 and L_2 intersect at exactly one point

L_1 and L_2 are coincident

L_1 and L_2 are parallel and distinct.

(see Fig 1.1) In the first case, the system has a unique solution corresponding to the single point of intersection of the two lines.

In the second case, the system has infinitely many solutions corresponding to the points lying on the same line

Finally in the third case, the system has no solution because the two lines do not intersect.

Let us illustrate each of these possibilities by considering some specific examples.

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